

TAIWO HENRY FADENI

APPLIED AI & SOFTWARE ENGINEER | LLM SYSTEMS • RAG & AGENT WORKFLOWS • DISTRIBUTED BACKENDS

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PROFESSIONAL SUMMARY

Applied AI & Software Engineer with an Electrical & Electronics Engineering foundation, specializing in production-grade LLM systems, evaluated RAG architectures, autonomous agent workflows, and high-throughput asynchronous backends. Experienced in building deterministic guardrails, human-in-the-loop controls, automated evaluation suites, and observable asynchronous backend pipelines. Proven track record of shipping end-to-end applications from distributed backend architecture to polished user interfaces.

TECHNICAL SKILLS

Programming Languages: Python, SQL (PostgreSQL, DuckDB, SQLite), JavaScript, MATLAB
AI & LLM Systems: LangGraph, LangChain, RAG Architecture, Vector Databases (Pinecone, ChromaDB), Embeddings, Semantic Caching, Prompt Engineering, Guardrails & Policy Gates, Structured Outputs (Pydantic), Human-in-the-Loop (HITL), Automated Evaluations (Golden Sets, Spider Benchmark)
Backend & Distributed Systems: FastAPI, WebSockets, RESTful APIs, Server-Sent Events (SSE), Celery, Redis (Pub/Sub, Caching, Sliding-Window Rate Limiting), AsyncIO, DuckDB, PostgreSQL, SQLite, Alembic, Pandas, PySpark
Frontend & UI Engineering: React, Next.js, State Management, Interactive Data Dashboards
DevOps, Cloud & Observability: AWS, Cloudflare, Docker, GitHub Actions (CI/CD), Git, Pytest (80%+ CI Coverage Gates), Prometheus, Grafana, Loki, Supabase, Vercel
Automation: n8n (self-hosted workflows, webhook orchestration, custom code nodes), event-driven pipelines, scheduled ETL, apps integration

APPLIED AI & BACKEND PROJECTS

Retrieval-Augmented Analytics Workspace (Text-to-SQL) | *FastAPI · DuckDB · sqlglot · Redis · SSE* 2026

- Natural-language analytics interface executing sandboxed, read-only analytical queries against DuckDB with streamed SSE explanations and dynamic charts.
- Built a 2-stage AST validation pipeline using sqlglot (enforcing table/column verification, write-query rejection, and injection safeguards) with self-correction retry logic.
- Automated evaluation harness across an 80-pair Golden Set (adapted from Spider), achieving **96% SQL validity**, **74% execution accuracy**, **61% failure self-correction**, and **~4.2s p95 latency**.

Self-Healing Microservices Monitor (Autonomous SRE Agent) | *LangGraph · ChromaDB · Prometheus · Postgres · React* 2026

- Incident-response agent integrating Prometheus Alertmanager webhooks, LangGraph multi-step diagnosis, and ChromaDB vector runbook retrieval.
- Designed a 4-condition deterministic policy gate (confidence ≥ 0.75 , allowlisted low-risk actions, impact checks, human approval routing) preventing destructive runaway executions with PostgreSQL audit trails.
- Scored **100% action and policy correctness** across simulated failure scenarios with a live React operator dashboard.

Distributed Realtime Chat Backend | *FastAPI · WebSockets · Redis Pub/Sub · SQLite · Docker* 2026

- Multi-room WebSocket backend scaling across workers using reference-counted Redis Pub/Sub channels (one channel per active room).
- JWT authentication at handshake, Redis-hash presence tracking with multi-device deduplication, and cursor-paginated message history (15 msgs/page); verified by cross-process integration tests for multi-worker delivery and connection fault isolation.

Autonomous Code Review Agent (GitHub App) | *Python · FastAPI · LangGraph · Celery · Redis · Docker* 2025

- Event-driven GitHub App agent that parses pull-request diffs, retrieves relevant file context, and publishes line-level inline reviews and commit statuses.
- Decoupled webhook intake from model inference using Celery workers for asynchronous job queuing, with HMAC-SHA256 webhook verification and Redis sliding-window rate limiting.
- Built a pluggable multi-provider LLM abstraction (OpenAI, Anthropic, Groq, Ollama) behind an **80% CI code coverage gate**.

EXPERIENCE

Freelance Applied AI & Software Engineer | *Independent Engineering & Consulting* Jan 2025 – Present (Remote)

- Delivered LLM systems, RAG workflows, agentic automation pipelines, and backend APIs across three engagements: a Series A logistics SaaS (Germany), an e-commerce operator (Nigeria), and a recruitment agency (Netherlands). Named references available on request.
- Automated lead qualification end-to-end with n8n and a classification agent, tripling qualified-lead volume (~45 → ~140/week) with no added headcount.

Electrical & Automation Engineering Intern | *Promasidor Nigeria Limited* May 2024 – Sep 2024 (Lagos, NG)

- Supported maintenance, diagnostic troubleshooting, and optimization of automated PLC-driven production-line systems in a high-volume FMCG manufacturing facility.
- Applied structured root-cause analysis (RCA) to resolve electrical and sensor faults, cutting recurring downtime; documented 5+ automation workflows and SOPs.

Student Research Assistant | *Communication Research Group & Control Systems Lab* 2023 – 2025 (OAU, NG)

- Co-developed an IoT-compatible telemetry station and simulated sensor-to-microcontroller data transmission using MATLAB and Simulink.
- Modeled Signal-to-Noise Ratio (SNR) across wireless configurations and contributed to smart metering prototypes for service-based tariff billing.

EDUCATION

B.Eng., Electrical & Electronics Engineering | *Obafemi Awolowo University* 2019 – 2025 (Ile-Ife, Nigeria)

Additional Training: Data Engineering Track (DataCamp, 2024–2025) • Software & Data Engineering (Data Epic, 2025) • Computational Thinking for Problem Solving (UPenn, 2022)